INCH-POUND

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SUPERSEDING
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PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING
TYPE 5865

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: TR, bandpass, frequency range 5,395 to 5,905 MHz, peak power rating 300 kw.

ABSOLUTE RATINGS:

| Parameter: | Incident power | Ebb | Altitude |
|---|---|---|---|
| Unit: | kw | V dc | feet |
| Maximum: | 300 | - - - | 10,000 |
| Minimum: | 4 | -700 | - - - |

PHYSICAL CHARACTERISTICS: See figure 1.

TEST CONDITIONS:

| Parameter: | tp1 | prr | Ii | Incident power | σ′ | F |
|---|---|---|---|---|---|---|
| Unit: | μs | pps | μA dc | kw | - - - | MHz |
| Tolerance: | ±0.15 | - - - | - - - | ±10% | Max | - - - |
| Test condition 1: | 1.0 | 1,000 | 100 | 4 | 1.1:1 | F3 |
| Test condition 2: | 1.0 | 1,000 | 100 | 70 | 1.1:1 | F3 |

| Frequency | | |
|---|---|---|
| F | MHz | ± (percent) |
| 1 | 5,395 | 0.1 |
| 2 | 5,450 | 0.1 |
| 3 | 5,650 | 5 |
| 4 | 5,825 | 0.1 |
| 5 | 5,905 | 0.1 |

GENERAL:

Qualification - Required.



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TABLE I. Testing and inspection.

| Inspection | Method MIL-STD-1311 | Notes | Test | Conditions | Symbol | Limits Min | Limits Max | Units |
|---|---|---|---|---|---|---|---|---|
| Qualification inspection | | | | | | | | |
| Degradation due to vibration | 4021 | - - - | - - - | | - - - | - - - | - - - | - - - |
| Flat leakage power | 4452 | - - - | 1 | | pf | - - - | 50 | mW |
| Spike leakage energy | 4452 | - - - | 1 | | Ws | - - - | 0.25 | erg |
| Conformance inspection, part 1 | | | | | | | | |
| Low-level VSWR | 4473 | - - - | - - - | $\sigma'$ = 1.05 max<br>F1<br>F2<br>F3 ±0.1 percent<br>F4<br>F5 | <br>$\sigma$<br>$\sigma$<br>$\sigma$<br>$\sigma$<br>$\sigma$ | <br>- - -<br>- - -<br>- - -<br>- - -<br>- - - | <br>1.9<br>1.4<br>1.4<br>1.4<br>1.9 | <br>- - -<br>- - -<br>- - -<br>- - -<br>- - - |
| Insertion loss | 4416 | - - - | - - - | F = F3; Ii = 0 | Li | - - - | 0.8 | dB |
| Ignitor interaction | 4421 | - - - | - - - | Ii = 100 μA dc | ΔLi | - - - | 0.2 | dB |
| Ignitor ignition time | 4401 | 1 | - - - | Ebb = -700 V dc<br>R = 4.0 MegΩ ±1 percent | t | - - - | 5 | sec |
| Ignitor voltage drop | 4406 | - - - | - - - | Ebb = -700 V dc<br>R = 4.0 MegΩ ±1 percent | Eid | 200 | 400 | V dc |
| Flat leakage power | 4452 | - - - | 2 | | pf | - - - | 50 | mw |
| Spike leakage energy | 4452 | - - - | 2 | | Ws | - - - | 0.25 | erg |
| Position of short | 4494 | 2 | 2 | | Distance | 0.070 | 0.100 | Inch |
| Arc loss | 4488 | - - - | 1 | | Loss | - - - | 0.80 | dB |
| Temperature cycling (nonoperating) | 1027 | - - - | - - - | | - - - | - - - | - - - | - - - |
| Conformance inspection, part 2 | | | | | | | | |
| Dielectric material strain | 4101 | 4 | - - - | | - - - | - - - | - - - | - - - |
| Recovery time | 4471 | - - - | 2 | | t | - - - | 10.0 | μs |
| Conformance inspection, part 3 | | | | | | | | |
| Life test | - - - | 3 | 2 | Group C;<br>Ebb = -700 V dc;<br>R = 4.0 MegΩ ±1 percent | t | 500 | - - - | hrs |

See footnotes at end of table I.



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TABLE I. Testing and inspection - Continued.

| Inspection | Method MIL-STD-1311 | Notes | Test | Conditions | Symbol | Limits Min | Limits Max | Units |
|---|---|---|---|---|---|---|---|---|
| Conformance inspection, part 3 -Continued | | | | | | | | |
| Life-test end points: | - - - | | | | | | | |
| Recovery time | 4471 | - - - | 2 | | t | - - - | 20 | μs |
| Flat leakage power | 4452 | - - - | 2 | | pf | - - - | 70 | mw |
| Insertion loss | 4416 | - - - | - - - | F = F3; Ii = 0 | Li | - - - | 0.8 | dB |
| Spike leakage energy | 4452 | - - - | 2 | | Ws | - - - | 0.3 | erg |
| Temperature cycling life test | 1027 | - - - | - - - | Group C; 10 cycles | - - - | - - - | - - - | - - - |

NOTES:

1. This test to be performed at the conclusion of the holding period.
2. With a metal plate shorting the line the position of the voltage standing wave minimum shall be determined. The metal plate shall be replaced by the tube and the position of the voltage standing wave minimum of the flat portion of the pulse shall be measured.
3. The ignitor current shall not be adjusted during life test. Life-test end points shall be measured using a fixed voltage and resistor.
4. After this test, the following requirement shall be met: Recovery time (Method 4471).
5. This specification sheet uses accept on zero defect sampling in accordance with MIL-PRF-1, table III.



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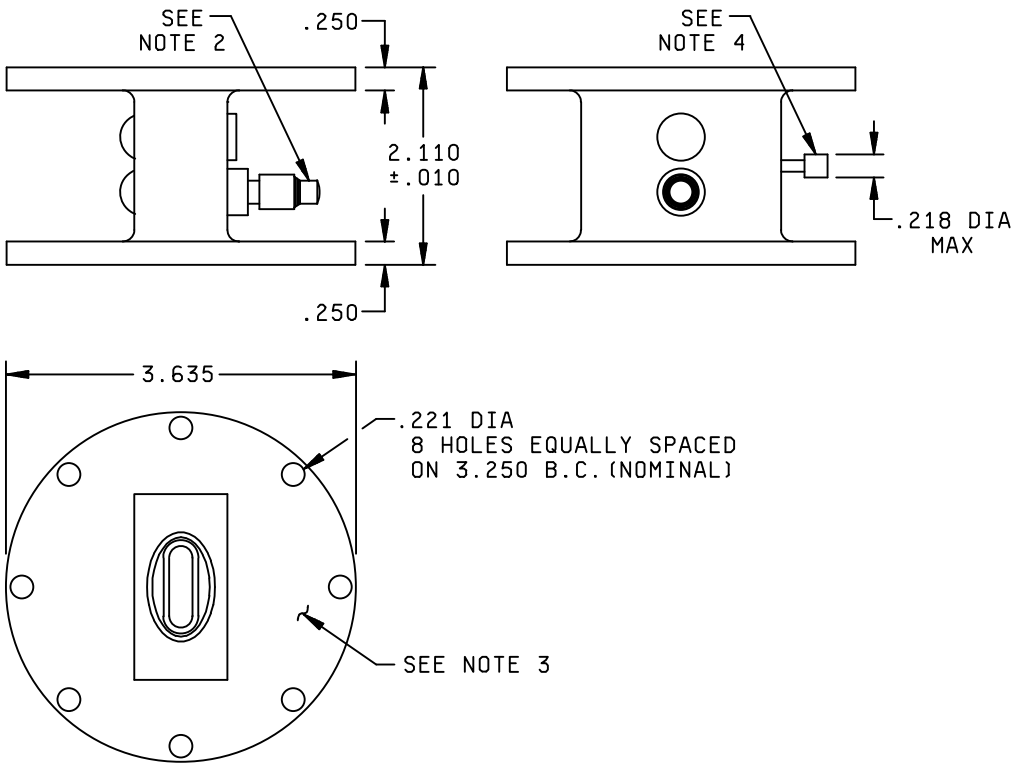


| Inches | mm |
|---|---|
| .010 | 0 0.25 |
| .218 | 0 5.54 |
| .221 | 0 5.61 |
| .250 | 0 6.35 |
| 2.110 | 53.59 |
| 3.250 | 82.55 |
| 3.625 | 92.08 |

NOTES:

1. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Skirted miniature cap C1-3. Cap not to extend beyond edge of flange.
3. Nickel plating or Rhodium flash over silver plating optional. If nickel plating is required, it is recommended that it be used only when other platings cannot meet performance requirements. The surface, referenced in this note, shall meet and enable the Gas Switching tube to meet all interface and performance requirements.
4. Exhaust tube not to extend beyond edge of flange.
5. Unless otherwise specified, tolerance is ±.016 (0.41 mm).

FIGURE 1. Outline drawing of electron tube type 5865.



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Referenced documents. In addition to MIL-PRF-1, this document references the following:
MIL-STD-1311

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2009-008)

Review activities:
Navy - AS, CG, MC, OS, SH
Air Force – 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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